A Brief Survey on Efficient Models for Qubits Encoding Using Quantum Machine Learning With Multi Level Time Series Data Classification

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Abstract

When it comes to realizing immediate quantum advantages in practical applications like machine learning and optimization techniques, numerous methods have emerged as a frontrunner technique. Such algorithms often involve quantum circuits for data encoding and the training of Quantum Neural Networks (QNNs) to minimize target functions when applied to tasks involving classical data. Quantum Machine Learning (QML) is a promising area of quantum computing because it may effectively handle difficult learning tasks by taking advantage of the high dimensional Hilbert space to train better representations from limited data. Despite QML's rising popularity, few academic works have addressed the language's security features. There has been a lot of buzz about the promising new discipline of quantum machine learning. Training a parameterized quantum circuit is a common practice in contemporary QML for performing data analysis on classical or quantum datasets. To make accurate predictions on an independent test dataset, state-of-the-art QML techniques first variationally optimize a parameterized quantum circuit using the training dataset. Quantum modeling in QML has also been shown to offer an exponential benefit in sample complexity. In the realms of pattern recognition, machine learning, and data mining, feature dimensionality reduction as a crucial link in the process has become one hot and tough spot. Most academics have gravitated towards this topic because of the difficulty of its studies. The goal of this research is to learn how to implement low loss in the procedure of feature dimension reduction, preserve the integrity of the original data, locate the most effective mapping, and obtain the best possible low-dimensional data. Preliminary findings suggest that QML models may offer advantages over classical models for classical data analysis. In this paper, a thorough investigation into generalization performance in QML is performed. This paper presents a brief survey on Qubits Encoding using Quantum Machine Learning models that help researchers to identify the limitations of the existing models and to design new efficient models for better performance levels.

Keywords: Quantum Machine Learning, Qubits, Data Encoding, Feature Dimensionality Reduction, Time Series Data, Generalization.

1. INTRODUCTION

In the disciplines of quantum artificial intelligence and computing, quantum machine learning represents a new and exciting area of study. As a result of their intrinsic parallel properties, quantum computers are thought to improve machine learning algorithms [1], showing quantum advantages over classical techniques. Notwithstanding their limits in quantum circuit breadth and depth, contemporary quantum devices can already show quantum advantages on some properly designed jobs [2]. The leading approach is the hybrid quantum-classical algorithms, also known as the variation quantum algorithms, and this is a crucial direction as it paves the way to explore quantum advantages with near-term quantum devices in real machine learning applications [3]. When training quantum neural networks, these methods typically rely on a classical optimizer [4]. Particularly, they delegate these relatively simple jobs to classical computers and provide the more complex ones to quantum computers.

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Applications in machine learning are not immune to the calls for quantum speedups that have been sparked by the rapid development of quantum computers [5]. One path is quantum-enhanced machine learning, which involves the use of quantum algorithms on traditional data sets. Many quantum machine learning algorithms have resulted from this line of enquiry [6]. Using fault-tolerant quantum computers has been shown to be advantageous over traditional methods in specific scenarios [7]. Finding realistic solutions for noisy intermediate-scale quantum (NISQ) devices is crucial due to the high expense of these technologies.

It is already a challenge on current quantum devices to encode classical information into quantum data, and this challenge will only increase in the near future. Parameterized quantum circuits (PQCs), such the empirically developed angle encoding, IQP encoding, etc., are among the most practical and widely used encoding schemes on NISQ devices [11-13]. The questions of how to pick these encoding schemes and whether or not they come with theoretical assurances are natural ones to explore [14]. The effect of such PQC-based encoding schemes on the efficiency of QNNs in quantum machine learning tasks must be thoroughly investigated. Parametric circuit elements include parameterized single-qubit rotations and entangling operations. To create entangled states, a collection of multi-qubit activities called entanglement operations is carried out between all of the qubits [15]. The following parametric single-qubit operations are used to seek the solution space. The term parametric layer is used to describe this set of entangling and single qubit rotating operations in QNN [16]. The optimum PQC design for QNN has been the subject of extensive study in recent years. In order to compare the relative merits of different PQC strategies [17], numerous characterizations have been provided. Supporters of these descriptors argue that the values of these descriptors have a strong correlation with the ability to train quantum circuits [18]. In practice, such descriptions could be helpful when deciding on the most appropriate PQC architecture for a certain QML use case.

The scale and complexity of time series data are two of their most distinguishing features. In addition to being wasteful, data mining from the original series is a time-consuming process. Thus, how to describe the time series data is the essential issue in the context of time series data mining. As a common feature reduction technology, time series recognition is widely used in time series data mining [19]. Some information will be lost in the transformation from high-dimensional to low-dimensional space using projections. The current challenge is to ideally preserve the fundamental qualities of the original data while obtaining meaningful reduction data from the high-dimensional data collection in order to meet the recognition accuracy and storage requirements [20]. Yet, in many real-world scenarios, locating and acquiring useful characteristics is more difficult. As a result, reducing dimensions has become a formidable challenge in the study of machine learning, data mining, and pattern recognition [20]. People are also paying more attention to dimension reduction as a field of basic research. Information dimension reduction has been addressed to a certain amount with numerous algorithms; however these solutions are not without flaws. Numerous researchers have put forth novel ideas that advance the study of pattern feature dimension reduction. This research presents a brief survey on efficient models for qubits encoding using quantum machine learning with multi level time series data classification that helps researchers to design novel methodologies with better performance levels.

2. LITERATURE SURVEY

In architecture of quantum bits (qubits) with nearest-neighbor interactions, Lechner et al. [1] described a way to parallelize this method for arbitrary all-to-all linked issue graphs. The optimization issue is encoded in a series of single-qubit operations, whereas interactions are problem-agnostic pairwise CNOT gates between nearest neighbours. This permits a parallelizable implementation in flat lattice-based quantum devices. This proposal builds on a lattice gauge model and adds new parameters and protocols to quantum approximate optimization algorithm in order to increase its effectiveness.

In order to be effective, many optimization algorithms and quantum machine learning methods rely on the creation of a quantum superposition. It's not easy to get ready an imperfect or non-uniform quantum superposition with desired characteristics. To construct a quantum superposition from a set of probability distributions, Anwer et al. [2] proposed a variational quantum circuit with n qubits that uses partial negation and controlled partial negation operators. The success of any quantum algorithm is critically dependent on the efficiency of the preparation procedure and the quality of the prepared superposition. Constructing the necessary quantum superposition can be done with the suggested method in O(n) steps and with good precision when compared to methods in the literature.

To make quantum secure direct communication (QSDC) enables direct, secret conversations over a quantum channel useful, however, users need to address its dependency on quantum memory, the immediacy with which communication can be disrupted by eavesdropping, and the low reliability of transmission induced by large qubit losses. For this reason, Sun et al. [3] proposed a new QSDC protocol based on the state-of-the-art coded single-photon DL04 QSDC protocol in order to address the outstanding difficulties. In particular, dynamic joint encryption and error-control (JEEC) coding allows us to forego the use of quantum memory while simultaneously deriving a highly precise estimate of the protocol's secrecy capacity.

The use of learning strategies to quantum devices has demonstrated the usefulness of exploiting quantum features.

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Data representation as quantum states is necessary for applying quantum computing to machine learning methods. An amplitude-encoded dataset, albeit more densely represented, is a more instructive input to a model. Amplitude encoding has two key benefits: improved classification accuracy and lower epoch-to-epoch learning variability. In this research, Sierra-Sosa et al. [4] evaluated the Quantum Convolutional Neural Network from TensorFlow Quantum, a hybrid quantum-classical network utilizing angle encoding, and a network of the own invention that employs amplitude encoding for enhanced state preparation.

Quantum stochastic neurons implements generalised linear models (QGLMs), use binary weights and activations to simplify their implementation. While previous work on training strategies for such models has focused on exhaustive search and bit-flip strategies reminiscent of single-neuron perceptrons, Nikoloska et al. [5] designed a stochastic variation optimization approach that allows for the joint training of quantum and traditional layers by means of stochastic gradient descent.

Author Name	**Year of Publication**	**Manuscript Title**	**Proposed Model**	**Limitations**
Lechner et al. [1]	2020	Quantum Approximate Optimization With Parallelizable Gates	The author described a way to parallelize this method for arbitrary all-to-all linked issue graphs. The optimization issue is encoded in a series of single-qubit operations, whereas interactions are problem-agnostic pairwise CNOT gates between nearest neighbours.	The considered models time complexity levels is high and the memory free protocol and code design contains complex operation that need to be reduced. The quantum classifiers performance can be enhanced by increasing the training samples for enhancing the accuracy levels.
Anwer et al. [2]	2022	Preparation of Quantum Superposition Using Partial Negation	The author proposed a variation quantum circuit with n qubits that uses partial negation and controlled partial negation operators. The success of any quantum algorithm is critically dependent on the efficiency of the preparation procedure and the quality of the prepared superposition.	
Sun et al. [3]	2020	Toward Practical Quantum Secure Direct Communication: A Quantum-Memory-Free Protocol and Code Design	The author proposed a new QSDC protocol based on the state-of-the-art coded single-photon DL04 QSDC protocol in order to address the outstanding difficulties. In particular, dynamic joint encryption and error-control (JEEC)	

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			coding allows us to forego the use of quantum memory while simultaneously deriving a highly precise estimate of the protocol's secrecy capacity.	
Sierra-Sosa et al. [4]	2020	TensorFlow Quantum: Impacts of Quantum State Preparation on Quantum Machine Learning Performance	The author evaluated the Quantum Convolutional Neural Network from TensorFlow Quantum, a hybrid quantum-classical network utilizing angle encoding, and a network of the own invention that employs amplitude encoding for enhanced state preparation.	
Nikoloska et al. [5]	2022	Training Hybrid Classical-Quantum Classifiers via Stochastic Variational Optimization	The author designed a stochastic variation optimization approach that allows for the joint training of quantum and traditional layers by means of stochastic gradient descent.	

Quantum Error Correction Codes (QECCs) are necessary in quantum processors because they increase the reliability of quantum logic gates. If the gate error probability is kept below a specific threshold, fault tolerant QECCs can improve the error rate of quantum processors. Below the depolarization probability bound, transversal QECCs provide a better error probability than an uncoded gate, and Cane et al. [6] quantified this bound in this paper. Steane's 7-bit QECC and a low-complexity repetition code are both characterized.

To combine these three distinct but supplementary types of features that are extracted to characterise global attributes of scenes, Wang et al. [7] proposed a discriminant association analysis-based feature-level fusion approach that maximises the correlation of associated characteristics across the two feature sets while also de-correlating characteristics that belong to different classes within each feature set. To further improve scene classification performance, a decision-level fusion version of information fusion is employed in the classification stage.

Image classification tasks with few training samples have been demonstrated to benefit more from the Bag-of-Deep-Visual-Words (BoDVW) model than the Convolutional Neural Network (CNN) model. BoDVW model relies heavily on a commercially available CNN model to extract deep features. In recent years, two deep feature extraction algorithms have been proposed. The first technique uses a deep-level fully-connected layer to transform intensively sampled multi-scale image patches into deep features. The second approach obtains deep features from the output of a deep-level convolutional layer or pooling layer. The second approach, on the other hand, is far more productive. However, its classification accuracy is lower than that of the previous approach. The reason for this is that the resulting receptive fields from the extraction of deep features using the second technique are uniform in size. Wang et al. [8] suggested augmenting the information gained from receptive fields of varying widths with deep features recovered by the second technique at low additional computing costs, hence enabling the BoDVW model to have high feature extraction efficiency and excellent classification accuracy.

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In this paper, Nelus et al. [9] discussed the potential privacy issues with acoustic sensor networks' with use of deep neural networks to extract features used for sound classification. It is performed by comparing two scenarios: one in which everyday activities at home are monitored using a single label, and another in which urban sounds are tagged using many labels. The author showed that in both circumstances, there is a considerable privacy risk for feature-intercept based speaker recognition attacks since feature representations used for sound classification also carry a significant quantity of speaker-dependent data. The field of visual tracking is becoming increasingly interested in discriminative correlation filters (DCF). In instance, a few recent experiments incorporate DCF into a Siamese network for image retrieval as a dedicated layer. Unfortunately, most of these efforts use shallow networks to learn target representations, which means they can't adapt to drastic changes in aesthetic style. In this paper, Liu et al. [10] created a unique Siamese network for correlation tracking by fusing high-level semantic data with low-level spatial detail features.

Author Name	**Year of Publication**	**Manuscript Title**	**Proposed Model**	**Limitations**
Cane et al. [6]	2020	Mitigation of Decoherence-Induced Quantum-Bit Errors and Quantum-Gate Errors Using Steane's Code	If the gate error probability is kept below a specific threshold, fault tolerant QECCs can improve the error rate of quantum processors. Below the depolarization probability bound, transversal QECCs provide a better error probability.	Qubits can be used for more than just quantum computing; they can also be applied to sensors. As a result, communications and encrypted networks pertaining to security can be delivered more quickly, and the safety of information stored inside the quantum field can be improved.
Wang et al. [7]	2020	Remote Sensing Scene Classification Using Heterogene Feature Extraction and Multi-Level Fusion	The author proposed a discriminant correlation analysis-based feature-level fusion approach that maximizes the correlation of corresponding features across the two feature sets while also de-correlating features that belong to different classes within each feature set in order to fuse these three distinct but complementary types of features that are extracted to characterize global attributes of scenes	The feature dimensionality reduction can be applied on the time series data for accurate classification levels.
Wang et al. [8]	2022	Improving Bag-of-Deep-Visual-Words Model via Combining Deep Features With Feature Difference Vectors	The author suggested augmenting the information gained from receptive fields of varying widths with deep features recovered by the second technique at low additional computing costs, hence enabling the	

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			BoDVW model to have high feature extraction efficiency and excellent classification accuracy.	
Nelus et al. [9]	2021	Privacy-Preserving Audio Classification Using Variational Information Feature Extraction	The author discussed the potential privacy issues with acoustic sensor networks' with use of deep neural networks to extract features used for sound classification	
Liu et al. [10]	2021	End-to-End Correlation Tracking With Enhanced Multi-Level Feature Fusion	The author created a unique Siamese network for correlation tracking by fusing high-level semantic data with low-level spatial detail features. To be more precise, the author developed a residual semantic embedding module to adaptively incorporate additional semantic information from high-level features to direct the feature fusion, hence introducing more semantic features into low-level features	

3. DISCUSSIONS

Quantum computing, with its powerful quantum tools for linear algebra, can enhance conventional machine learning methods. As linear algebra is the foundation of ML, it stands to reason that ML methods would benefit from incorporating quantum computing techniques. Training machine learning algorithms is similar in many ways to tuning gate parameters to change qubit states. The QML algorithm for ML problems can be realized as a series of quantum gate operations in a quantum circuit. The performance of massive data processing can be enhanced in high-stakes, real-world applications by applying quantum machine learning techniques. Unfortunately, there is a limit to how many qubits may be used in a quantum computer at the moment. While the current generation of quantum computers offers several advantages, they are limited by environmental noise. Qubits' interactions with their environment make it possible for the data they store to be lost. That's why it's hard to solve ML problems with the noisy intermediate-scale quantum devices of today. It is critical that QML algorithms be developed that are compatible with the limitations of existing quantum computers. This research work focuses on the intersection of quantum computing with classical machine learning, a field known as Quantum Machine Learning for Big Data Analytics, with the goal of enhancing supervised learning in classical big data. Quantum computational methods are investigated for their potential to enhance conventional machine learning algorithms used in large data analysis.

Feature selection is a technique of selecting feature subsets that are used in model construction. It is also known as variable selection or feature subset selection. The model can be simplified so that it is easier to interpret for researchers, the run time can be reduced, the curse of dimensionality can be avoided, and generalization can be improved by removing

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excessive fitting. Most importantly, the data must have numerous duplicate or related features that may be removed without losing crucial details in order to employ feature selection methods. Because a related feature can become redundant when there are already other related features present, it is not the same thing as a redundant feature. Choosing which features to use can sometimes be framed as an optimization challenge involving a search. It is shown that finding the smallest possible subset of features to search for is an NP issue. The best answer can only be found after a thorough search. The computational cost of conducting an extensive search is typically high. The best feature subset can be found by exploring all feasible M-feature combinations of the initial set of N features. Due to this combination explosion, the quantity of calculation required grows exponentially as the total number of features does. This means that in most cases, it is impossible to do a search-optimized thorough search.

This paper presents a brief survey on Qubits Encoding using Quantum Machine Learning models that help researchers to identify the limitations of the existing models.In order to ensure the safety of the data, this research suggested developing a model for quantum-based time series data processing that encodes the data into qubits using parameterized quantum gates. In order to create quantum representations of information by measuring qubits in a quantum circuit, it is necessary to develop Quantum measurement based characteristics extracted from time series data. The goal is to design a model and to train a model using time series data and a feature dimension reduction technique that includes feature vector generation and to design a machine learning model to create a multi-level categorization system based on a vector of features and to improve classification accuracy by combining an optimization model with a multi-level approach.

4. CONCLUSION

There has been a lot of buzz about the promising new discipline of quantum machine learning. Training a parameterized quantum circuit is a common practice in contemporary QML for performing data analysis on classical or quantum datasets. Preliminary findings suggest that QML models may offer advantages over classical models for classical data analysis. For the purpose of processing quantum data, QML models have been shown to offer an exponential benefit in sample complexity. Learning a unitary that is represented by a polynomial-depth quantum circuit is a more realistic scenario than considering arbitrary unitaries. This subset of unitaries is exponentially smaller than that of arbitrary unitaries, and corresponds to the ones that can be efficiently performed on a quantum computer. In a broader sense, one may relate the required size of the training data set N for generalization to the number of parameters in the QML model. Using generalization error as a dynamic number those changes as the optimization progresses is an even more generic approach. In this research, a wide range of dimensionality reduction and classification methods are analyzed to see which ones best enhance the learning performance of algorithms and, by extension, the predictive accuracy of various classifiers. Some dimensionality reduction algorithms focus on removing only redundant features, while others focus on eliminating all irrelevant features. Depending on the task at hand, these algorithms either create a new set of features that are similar to the original set of features or create a feature subset consisting of features from the original set. This paper presents a brief survey on Qubits Encoding using Quantum Machine Learning models that help researchers to identify the limitations of the existing models. This literature survey helps numerous researchers to identify the limitations of QML and feature extraction and selection in time series data to design new solutions for better performance levels.

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